

1. Administrative & Study Identification

**Full Study Title:** A Phase III, Multicenter, Open-Label Study to Evaluate the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Emicizumab Prophylaxis in Patients With Type 3 Von Willebrand Disease **Summary Title:** A Study to Assess the Efficacy and Safety of Emicizumab in Participants With Type 3 Von Willebrand Disease **Short Title/Acronym:** WILL-EMI **Posting ID:** 145202570946529316 **Protocol Number:** WP45338 (2024-515622-80-00) **NCT Number:** NCT06998524 **Trial Status:** RECRUITING **Sponsor/Company Name:** Roche (Hoffmann-La Roche) **Investigational Product:** Emicizumab (Hemlibra) **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

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2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To evaluate the efficacy of emicizumab prophylaxis in reducing the Annualized Bleed Rate (ABR) for treated bleeds in the randomized arms for participants with Type 3 von Willebrand disease (VWD) compared to standard of care (SOC) on-demand therapy. **Secondary Objectives:** To evaluate the ABR for all bleeds in the randomized arms, treated spontaneous bleeds in the randomized arms, and treated joint bleeds in the randomized arms; and to assess the long-term safety, pharmacokinetics, and pharmacodynamics of emicizumab.

**2.2 Background & Rationale** Type 3 VWD is the rarest and most severe form of VWD, characterized by a near-total absence of von Willebrand factor, leading to severe spontaneous and provoked bleeding episodes. Current preventative treatments require frequent intravenous (IV) infusions, imposing a heavy burden on patients. Emicizumab, a bispecific antibody administered subcutaneously, has shown efficacy in hemophilia A and is hypothesized to prevent bleeding and improve the quality of life for patients with Type 3 VWD.

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3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional **Control Type:** Active Comparator (SOC on-demand therapy) and intra-participant historical control (NIS WP45335, NCT06883240) **Blinding:** Open-label **Randomization:** Randomized (Arms A and B for prior on-demand patients) and Non-randomized (Arm C for prior prophylaxis patients)

**3.2 Planned Enrollment & Duration Target** **\$N\$:** 75 participants  
**Number of Sites:** Multicenter (Global: United States, Europe, Canada, Japan, RoW) **Estimated Study Start:** 27-Jun-2025 **Estimated Primary Completion:** 31-Dec-2026

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## **4. Subject Selection (Eligibility Criteria)**

**4.1 Inclusion Criteria** 1. Confirmed diagnosis of Type 3 von Willebrand disease (VWD), based on medical records. 2. Preexisting medical record verifying the status of von Willebrand factor (VWF) inhibitor (positive or negative, including titer if available). 3. Adequate hematologic, hepatic, and renal function. 4. For participants of childbearing potential: agreement to remain abstinent or adhere to the contraception requirements. 5. **Arms A & B:** Age  $\geq 1$  month at the time of signing Informed Consent/Assent Form. Documented previous use of intermittent (less than once a week) on-demand SOC therapy for VWD. Having  $\geq 2$  treated bleeds (except menstrual bleeds) with factor concentrate within 24 weeks prior to enrollment. 6. **Arm C:** Age  $\geq 2$  years at the time of signing Informed Consent/Assent Form. Documented and confirmed previous use of SOC prophylactic therapy for VWD (1-3 times weekly, as per prescribed dose) as described in the eligibility of Study WP45335. Have completed all study requirements as defined in the WP45335 protocol for at least 24 weeks.

**4.2 Exclusion Criteria** 1. Inherited or acquired bleeding disorder other than Congenital Type 3 VWD. 2. History of gastrointestinal bleeding within 18 months prior to enrollment, or any previous diagnosis of angiodysplasia. 3. History of intracranial hemorrhage. 4. Previous or current treatment for thromboembolic disease or signs of thromboembolic disease. 5. Other conditions (e.g., certain autoimmune diseases) that may increase risk of bleeding or thrombosis. 6. History of clinically significant hypersensitivity associated with monoclonal antibody therapies or components of the emicizumab injection. 7. Use of systemic immunomodulators (e.g., interferon) at enrollment or planned use during the study, with the exception of anti-retroviral therapy.

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## **5. Intervention & Treatment Plan**

**5.1 Investigational Regimen Dosage/Frequency:** Emicizumab 3 mg/kg administered once a week (QW) for the first 4 weeks as loading doses, followed by a maintenance dose of 3 mg/kg once every 2 weeks (Q2W). **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** 24-week primary assessment period, with an opportunity to continue emicizumab prophylaxis in an extension period.

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard of care treatments (e.g., VWF concentrates) used on-demand to manage

breakthrough bleeds according to local labeling. **Prohibited:** Systemic immunomodulators (excluding anti-retroviral therapy) and specific concurrent investigational drugs.

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## 6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---  
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, verification of VWF inhibitor status | |  
**Baseline** | Day 0 | Randomization, First SC loading dose (Arms A/C) or SOC continuation (Arm B) | | **Interim** | Weeks 4, 12 | Completion of loading phase, Safety Labs, ongoing Bleed Diary review | | **End of Study** | Week 24 | Primary ABR evaluation, transition to extension period or safety follow-up |

*(Note: Patients discontinuing emicizumab will undergo a Safety Follow-up Visit 6 months after their final dose)*

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Annualized Bleed Rate (ABR) for Treated Bleeds in the Randomized Arms. **Measurement Method:** Evaluated through patient/caregiver-reported bleeding diaries and investigator clinical assessment.

**7.2 Secondary & Exploratory Endpoints** \* **Endpoint 1:** ABR for All Bleeds in the Randomized Arms, ABR for Treated Spontaneous Bleeds in the Randomized Arms, and ABR for Treated Joint Bleeds in the Randomized Arms. \* **Endpoint 2:** Intra-participant comparison of the ABR for Treated Bleeds with Prophylactic Emicizumab versus Prophylactic SOC from the preceding non-interventional study (WP45335, NCT06883240).

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Standard collection via scheduled visits and bleed diaries, with a specific focus on thromboembolic events and hypersensitivity site reactions. **Serious Adverse Events (SAEs):** Routine expedited reporting timeline (24 hours). **Data Monitoring Committee (DMC):** Independent safety monitoring panel evaluating bleeding risks and thrombotic events throughout the trial.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Intent-to-Treat (ITT) population for primary efficacy comparisons (Arm A vs. Arm B). Safety Set for all participants receiving  $\geq 1$  dose of emicizumab. **Sample**

**Size Justification:** \$N = 75\$, powered to detect a statistically significant reduction in ABR between emicizumab prophylaxis and standard on-demand care. **Handling of Missing Data:** Adjustments utilizing negative binomial regression models corresponding to the exact observation time (person-years) for missing entries.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

**Monitoring Plan:** Continuous site monitoring (remote and onsite) focusing on the accuracy of bleed reporting and SC administration compliance. **Audit Trail:** Secure eCRF locking and adherence tracking for bleed diaries and concomitant factor concentrate usage.

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## 11. Study Identifiers & Governance

**Primary ID:** WP45338 **Registry Number:** NCT06998524 **Sponsor/Lead Organization:** Roche **Study Title:** WILL-EMI Study **Official Regulatory Title:** A Phase III, Multicenter, Open-Label Study to Evaluate the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Emicizumab Prophylaxis in Patients With Type 3 Von Willebrand Disease

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## 12. Clinical Framework (Type 3 VWD Specific)

**Primary Condition:** Type 3 Von Willebrand Disease (VWD) **Diagnosis Threshold:** Confirmed Type 3 VWD based on medical records **Medical Methodology:** Bispecific Factor IXa- and Factor X-directed antibody Prophylaxis **Optimization Target:** Bleed prevention and ABR reduction

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## 13. Study Procedures & Methodology

**Management Pathway:** Transition from SOC factor concentrates to SC Emicizumab regimen **Education Protocol (HCP):** Emicizumab administration, bleed evaluation, and rescue therapy training **Education Protocol (Patient):** Subcutaneous injection self-administration and rigorous diary recording **Quality Audit Mechanism:** Continuous reconciliation of dispensed factor concentrates against patient bleed diaries

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## 14. Observation & Follow-Up Schedule

**Total Duration:** 24-week initial evaluation (with extension phase and 6-month safety follow-up post-treatment) **Onsite Visit**

**Cadence:** Baseline, Week 4, Week 12, Week 24 **Remote Monitoring**  
**Frequency:** Intermittent review of electronic bleeding diaries  
**Checklist Review Interval:** Regular tracking of breakthrough bleed treatments and spontaneous bleed events

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**15. Sign-off & Approval**

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|---------------------------------|--------|-----------|--------------------|------|-------------|------|------|
| Role                            | Name   | Signature | Date               | :--- | :---        | :--- | :--- |
| <b>Principal Investigator</b>   | [Name] | _____     | <b>07-Apr-2026</b> |      |             |      |      |
| <b>Clinical Operations Lead</b> | [Name] | _____     | <b>07-Apr-2026</b> |      | <b>Lead</b> |      |      |
| <b>Statistician</b>             | [Name] | _____     | 07-Apr-2026        |      |             |      |      |